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Digital Signal Processing 1 Laboratory

Final Project Report

Section: C1 Group: 04

BIOMETRIC AUTHENTICATION SYSTEM FOR CLASSROOM MANAGEMENT THROUGH VOICE RECOGNITION

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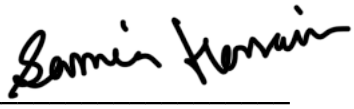
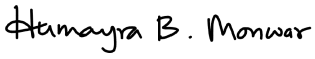
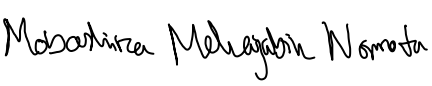
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1 Abstract

Secure authentication and access control are crucial aspects of modern classroom management systems, ensuring the protection of sensitive educational resources and preventing unauthorized access. Traditional methods such as passwords and identification cards pose security risks and can be inconvenient or easily compromised. Biometric authentication systems, specifically those based on voice recognition, offer a more robust and user-friendly solution for classroom environments.
This project aims to develop a biometric authentication system for classroom management through voice recognition technology. The proposed system utilizes advanced signal processing techniques, including Mel-Frequency Cepstral Coefficients (MFCC) for efficient feature extraction from voice samples. Machine learning algorithms, such as k-Nearest Neighbors (KNN) is employed for speaker modeling and classification.
The system is designed to accommodate text-dependent speaker recognition. Robust audio preprocessing methods, including noise reduction and voice activity detection, are incorporated to improve the system's performance in real-world classroom environments.
Through extensive experimentation and evaluation, the developed system demonstrates high accuracy in speaker recognition. Additionally, the project addresses potential privacy concerns and ethical considerations related to the collection and storage of biometric data, ensuring compliance with relevant regulations and best practices.
The proposed biometric authentication system for classroom management through voice recognition offers a secure, convenient, and user-friendly solution for educational institutions. By enhancing access control and ensuring the protection of sensitive resources, this project contributes to the advancement of secure and efficient classroom management practices.

2 Introduction

Adherence to manual attendance methods—like paper-based tracking or roll calls in person—highlights inefficiencies and the ease with which students can provide proxy attendance. By leveraging the distinctive qualities of human voice signals, speaker recognition technology is proposed as a replacement for traditional biometric verification methods, providing a solution that offers accuracy and efficiency. In order to complement an application created for easy classroom attendance tracking, this early-stage project aims to gather student voice samples and compile them into a single database. By reducing wasted class time and preventing proxy attendance, this creative technique significantly improves attendance management procedures.

3 Design

3.1 Problem Formulation (PO(b))

The project aims to develop a Biometric Authentication System for Classroom Management through Voice Recognition, addressing the need for efficient and secure attendance tracking in educational settings. Drawing on principles from electrical and electronic engineering, the system formulates algorithms for signal processing and pattern recognition to effectively capture and analyze voice samples, incorporating techniques such as filtering, feature extraction, and machine learning-based identification. Extensive research literature in digital signal processing, pattern recognition, database management, and ethical considerations

informs the development process, ensuring compliance with relevant standards and regulations governing biometric data usage. Data interpretation involves analyzing voice samples, extracted features, and attendance records to generate reliable authentication outcomes and attendance summaries. The project navigates complex engineering challenges, including variability in voice data, trade-offs between security and usability, scalability concerns, and seamless integration into existing workflows. By addressing these challenges innovatively, the system aims to provide educational institutions with a robust, scalable, and user-friendly solution for classroom management, enhancing efficiency and security in attendance tracking while adhering to ethical and legal standards.

3.1.1 Identification of Scope

The scope of the project encompasses the development of a Biometric Authentication System for Classroom Management through Voice Recognition. This includes defining the system's objectives, functionalities, and target users. Key aspects of the scope involve determining the system's capability to accurately identify individuals based on their voice patterns, integrate seamlessly into existing classroom management workflows, and ensure data privacy and security.

3.1.2 Literature Review

Speaker recognition is a biometric technology that uses an individual's voice characteristics to verify or identify them. It has gained significant attention in recent years due to its potential applications in various domains, including security systems, human-computer interaction, and classroom management.

Farah and Shamim (2013) implemented a speaker recognition system using Mel-Frequency Cepstrum Coefficients (MFCC) and Linear Prediction Coding (LPC) for feature extraction, and Vector Quantization (VQ) for speaker classification. Their study compared the efficiency of MFCC and LPC, as well as text-dependent and text-independent speaker identification. The results showed that MFCC outperformed LPC, and text-dependent recognition achieved higher accuracy than text-independent recognition. Additionally, they found that the presence of noise and pitch alteration negatively affected the system's accuracy.

Subba Rao et al. (2020) explored the use of the k-Nearest Neighbor (kNN) algorithm for speaker recognition. They employed various audio preprocessing techniques, such as trimming, noise reduction, and vocal enhancement, to improve the quality of recorded audio samples. The study evaluated the performance of kNN using MFCC coefficients, MFCC delta coefficients, and MFCC delta-delta coefficients. The results indicated that the combination of kNN and MFCC coefficients achieved the highest classification accuracy among the tested configurations.

In the context of classroom management, speaker recognition systems can be used to authenticate students and instructors, ensuring secure access to educational resources and preventing unauthorized entry. Billson Mokgonyane et al. (2019) developed an automatic speaker recognition system based on machine learning algorithms, specifically targeting under-resourced languages. Their study highlighted the potential of such systems in enhancing security and accessibility in educational settings.

Several challenges and considerations arise when implementing speaker recognition systems for classroom management. Factors such as background noise, variations in speaking styles, and the need for text-independent recognition (as students may not always recite predetermined phrases) can affect the system's performance. Additionally, privacy concerns and ethical considerations related to the collection and storage of biometric data should be addressed.

To develop an effective biometric authentication system for classroom management through

voice recognition, it is crucial to explore robust feature extraction techniques, advanced machine learning algorithms, and efficient audio preprocessing methods. Additionally, integrating speaker recognition with existing classroom management systems and addressing potential security and privacy concerns are essential aspects to be considered.

3.1.3 Formulation of Problem

The problem formulation involves identifying the challenges and requirements inherent in developing a voice-based authentication system for classroom management. This includes addressing issues such as variability in voice data, ambient noise, scalability concerns, and the need for real-time authentication. Formulating the problem entails defining clear objectives, constraints, and success criteria for the project, as well as considering ethical and legal implications surrounding the collection and usage of biometric data.

3.1.4 Analysis

The analysis phase involves a comprehensive examination of the technical, operational, and societal factors relevant to the development and implementation of the authentication system. This includes analyzing existing literature, standards, and regulations pertaining to voice recognition, biometric authentication, and data privacy. Furthermore, the analysis entails evaluating the feasibility of various technical approaches, assessing potential risks and constraints, and identifying opportunities for innovation and improvement. Through rigorous analysis, the project aims to develop a well-informed strategy for designing, implementing, and deploying the biometric authentication system effectively within educational environments.

3.2 Design Method (PO(a))

We followed the procedures mentioned below to implement this project:

A. Database and Dataset

Our project involved collecting 15 distinct voice samples from each student, correlating with their respective student IDs, resulting in a comprehensive database comprising 150 samples from 10 students. Each audio sample, with a duration of 3 seconds and recorded in mono channel format, has a sampling frequency of 16000Hz. These recordings were captured under various everyday conditions to ensure the robustness of our model, enabling it to accurately recognize voices within the dynamic environment of a classroom.

B. Filtering Noise

It is well known that a significant proportion of a voice signal's frequency components are low frequency, with the majority falling well inside 3000 Hz. Consequently, we employed a noise cancelling low pass filter with a 5th order cutoff frequency of 3000 Hz to eliminate high frequency colored noise from our voice samples.

C. Removing silent part

Removing silent parts involves preprocessing the audio data to eliminate periods of silence or background noise that may not contribute meaningful information for voice recognition. This process is essential for improving the quality of the audio signals and enhancing the accuracy of subsequent feature extraction and classification steps.

In our code, the removal of silent parts is implemented using a custom function. This function analyzes the audio signal and identifies segments with low energy levels, which are indicative of silence or background noise. These segments are then

removed, and the remaining portions of the audio signal, containing the desired speech segments, are retained for further processing.

The implementation typically involves setting a threshold energy level below which audio segments are considered silent. Any segments with energy levels below this threshold are removed or attenuated, while segments with energy levels above the threshold are retained.

D. Standardization and Normalization

In our project, voice signals were meticulously standardized and normalized to enhance the consistency and reliability of our speaker recognition system. Standardization involved adjusting the audio recordings to a uniform format and duration, ensuring each sample maintained a consistent quality and structure. Normalization entailed adjusting the volume levels across all samples to a common baseline, mitigating discrepancies that could affect the model's ability to accurately identify and analyze voice characteristics. These processes were crucial for reducing variability and improving the model's performance, allowing it to more effectively distinguish between individual voices under the diverse acoustic conditions encountered in a classroom setting.

E. Feature Extraction

Feature extraction is a critical step in voice recognition systems, where relevant characteristics or features are extracted from the audio signals to represent the unique aspects of an individual's voice.

In our code, features such as Mel-Frequency Cepstral Coefficients (MFCCs), pitch, spectral centroid, zero-crossing rate, and short-time energy have been extracted. MFCCs are widely used in speech processing due to their effectiveness in capturing spectral characteristics while minimizing the effect of noise and other irrelevant factors. Pitch represents the fundamental frequency of the voice, while spectral centroid measures the center of mass of the frequency spectrum. Zero-crossing rate indicates the rate at which the audio signal changes sign, providing information about its temporal characteristics. Short-time energy quantifies the signal's energy within short time intervals.

The implementation involves segmenting the audio signal into frames, typically using a windowing technique, and computing the desired features for each frame. These features are then aggregated over time to create a feature vector representing the entire audio signal, which serves as input to machine learning models for training and testing.

F. Data splitting

We split our dataset, reserving 20% of the data for testing and 80% for training.

In building our model, we chose the k-Nearest Neighbors (KNN) algorithm for its simplicity and effectiveness in classification tasks. Theoretically KNN is the best classifier if we can predict the required number of K neighbors in KNN (K Nearest Neighbors). So, we used KNN as our classifier. KNN computes an input’s distance from each vector in the dataset and finds the k closest vectors. The most anticipated class from these k vectors is returned. This simple strategy works in many machine learning algorithms. The k value estimation and distance metric are crucial to this approach. The elbow approach is used to estimate k until the accuracy no longer improves. We can use different distance for KNN. KNN model selects the right number of neighbors (k) and the distance metric used to measure similarity between samples. We explored a range of k values and various distance metrics, including Euclidean,cityblock, searman, Chebyshev, Minkowski, and others, to find the best combination for our dataset. Best parameters were found by hyperparameter tuning

and 5 fold cross validation was performed.

Here for our code, the algorithm is:

- **Euclidean Distance Calculation:** Compute the distance between real-time and stored features for similarity assessment
- **K-Value Selection:** Select the three nearest neighbors (k=3) for their minimal distances.
- **Majority Voting:** Determine the tentative ID based on the majority vote among the k-selected IDs.
- **Feature-Wide Application:** Repeat the process across all 43 extracted features.
- **Tabulation of Predictions:** Store each feature's predicted labels in a structured table for analysis
- **Final Label Determination:** Assign the final student ID based on the most frequently occurring label in the table.

After testing, we get the Validation Accuracy: 95.00% and Test Accuracy: 93.33%.

G. Attendance Database:

- **Performance Tracking:** Visual representation through bar charts provides an at-a-glance understanding of each student's attendance record, enabling easy tracking of participation trends over time.
- **Engagement Insights:** Attendance percentages highlight engagement levels, identifying students who may need additional support or recognition for consistent attendance.
- **Data-Driven Decisions:** Facilitates informed decision-making for faculty on curriculum pacing and intervention strategies, based on attendance patterns.

H. Real time testing

In our code, real-time testing is a crucial step in the Biometric Authentication System for Classroom Management through Voice Recognition. Here's how it is implemented:

- **Audio Capture:** The system prompts the user to start speaking, initiating the recording of audio data through a microphone in real-time using the **audiorecorder** function.
- **Preprocessing:** The captured audio undergoes preprocessing steps, including filtering and removal of silent parts, achieved through custom functions. Filtering ensures that the audio is clean and free from noise, while the removal of silent parts eliminates irrelevant segments.
- **Feature Extraction:** After preprocessing, feature extraction techniques such as Mel-Frequency Cepstral Coefficients (MFCCs), pitch, spectral centroid, zero-crossing rate, and short-time energy are applied to the real-time audio signal. This is done using an audioFeatureExtractor object initialized with appropriate parameters to extract the desired features.
- **Normalization:** The extracted features are then normalized using the mean and standard deviation computed during training. This normalization ensures that the features have consistent scales and distributions, facilitating accurate comparison and classification. The normalization process is crucial for maintaining consistency between the training and testing phases.
- **Classification:** The normalized features are fed into the trained machine learning model, implemented using algorithms such as k-nearest neighbors (k-NN), to predict the identity of the speaker in real-time. The model predicts the speaker's identity based on the similarity between the real-time features and the features extracted during training.

- **Result Interpretation:** The final predicted label or authentication result is obtained from the classification output. This result determines the speaker's identity or authentication status, which can be used for attendance tracking or access control purposes.

3.3Full Source Code of Firmware

```
clc;
clear;
close all;
warning off;
choice = 0;
position = 5;
while choice ~= position
choice = menu('Voice Based Attendance System', 'Add Voice to the Database', 'Train Model',
'Identify Student', 'Database Information', 'Exit');
%% Data collection
if choice == 1
clc;
id=input('Enter your student ID (7 digits)==> ','s');
maindir=['C:\Users\GOAT\humayra\buet\3-1\eee 312\project final\data\' id];
mkdir(maindir);
fs=16000; %sampling frequency
recobj=audiorecorder(fs,16,1);
p=1;
repeat='y';
while repeat~='n'
clc;
disp('press any key to start recording');
pause
pause(1);
disp('Speak Now!');
recordblocking(recobj,3);
y=getaudiodata(recobj);
f=[id '_' num2str(p) '.wav'];
DIR=[maindir '\' f];
audiowrite(DIR,y,fs);
fprintf('recording done %d times\n',p);
repeat=input('Do another recording?(n==quit)==> ','s');
p=p+1;
end
%% Training
elseif choice == 2
clc;
warning off;
%% Filter
original_data_dir = "C:\Users\GOAT\humayra\buet\3-1\eee 312\project final\data";
filtered_data_dir = "C:\Users\GOAT\humayra\buet\3-1\eee 312\project final\filtered_data";
if ~exist(filtered_data_dir, 'dir')
mkdir(filtered_data_dir);
end
% Filter specifications
Fs = 16000; % Sampling frequency
Fc = 3000; % Cut-off frequency for the low pass filter
order = 5; % Order of the filter
% Processing each student's folder
studentsFolders = dir(fullfile(original_data_dir, '*'));
for i = 1:length(studentsFolders)
if studentsFolders(i).isdir && ~strcmp(studentsFolders(i).name, '.') &&
~strcmp(studentsFolders(i).name, '..')
```

```
studentID = studentsFolders(i).name;
studentInputDir = fullfile(original_data_dir, studentID);
studentOutputDir = fullfile(filtered_data_dir, studentID);
if ~exist(studentOutputDir, 'dir')
mkdir(studentOutputDir);
end
voiceSamples = dir(fullfile(studentInputDir, '*.wav'));
% disp(['Processing student ID: ', studentID]);
for j = 1:length(voiceSamples)
fileName = voiceSamples(j).name;
[y, Fs] = audioread(fullfile(studentInputDir, fileName));
d=designfilt('lowpassfir','FilterOrder',order,'CutoffFrequency',Fc,'SampleRate',Fs);
filteredSignal=filtfilt(d,y);
audiowrite(fullfile(studentOutputDir, fileName), filteredSignal, Fs);
end
end
end
%% VAD
vad_dir = "C:\Users\GOAT\humayra\buet\3-1\eee 312\project final\vad_data";
if ~exist(vad_dir, 'dir')
mkdir(vad_dir);
end
% Processing each student's folder
studentsFolders = dir(fullfile(filtered_data_dir, '*'));
for i = 1:length(studentsFolders)
if studentsFolders(i).isdir && ~strcmp(studentsFolders(i).name, '.') &&
~strcmp(studentsFolders(i).name, '..')
studentID = studentsFolders(i).name;
studentInputDir = fullfile(filtered_data_dir, studentID);
studentOutputDir = fullfile(vad_dir, studentID);
if ~exist(studentOutputDir, 'dir')
mkdir(studentOutputDir);
end
voiceSamples = dir(fullfile(studentInputDir, '*.wav'));
% disp(['Processing student ID: ', studentID]);
for j = 1:length(voiceSamples)
fileName = voiceSamples(j).name;
[y, Fs] = audioread(fullfile(studentInputDir, fileName));
new_signal = removing_silentpart(y, Fs);
audiowrite(fullfile(studentOutputDir, fileName), new_signal, Fs);
end
end
end
%% Standardization and Normalization
standardized_dir = "C:\Users\GOAT\humayra\buet\3-1\eee 312\project
final\standardized_data";
if ~exist(standardized_dir, 'dir')
mkdir(standardized_dir);
end
% Processing each student's folder
studentsFolders = dir(fullfile(vad_dir, '*'));
for i = 1:length(studentsFolders)
if studentsFolders(i).isdir && ~strcmp(studentsFolders(i).name, '.') &&
~strcmp(studentsFolders(i).name, '..')
studentID = studentsFolders(i).name;
studentInputDir = fullfile(vad_dir, studentID);
studentOutputDir = fullfile(standardized_dir, studentID);
if ~exist(studentOutputDir, 'dir')
mkdir(studentOutputDir);
end
voiceSamples = dir(fullfile(studentInputDir, '*.wav'));
% disp(['Processing student ID: ', studentID]);
```

```
for j = 1:length(voiceSamples)
fileName = voiceSamples(j).name;
[y, Fs] = audioread(fullfile(studentInputDir, fileName));
y = y - mean(y);
y = y / std(y);
audiowrite(fullfile(studentOutputDir, fileName), y, Fs);
end
end
end
%% Adding to the Database
if ~exist('id_database.dat', 'file')
data = {};
sound_number = 0;
speaker_num = 0;
else
load('id_database.dat', '-mat');
end
studentsFolders = dir(fullfile(standardized_dir, '*'));
for i = 1:length(studentsFolders)
if studentsFolders(i).isdir && ~strcmp(studentsFolders(i).name, '.') &&
~strcmp(studentsFolders(i).name, '..')
studentID = studentsFolders(i).name;
studentDir = fullfile(standardized_dir, studentID);
voiceSamples = dir(fullfile(studentDir, '*.wav'));
for j = 1:length(voiceSamples)
fileName = voiceSamples(j).name;
filePath = fullfile(studentDir, fileName);
[y, Fs] = audioread(filePath);
sound_number = sound_number + 1;
speaker_num = speaker_num + 1;
data{sound_number, 1} = y; % The audio data
data{sound_number, 2} = studentID; % Student ID as 'class'
data{sound_number, 3} = 'Microphone'; % Source of the recording
data{sound_number, 4} = num2str(speaker_num); % Unique speaker number
end
end
end
%% Feature Extraction
features_dir = "C:\Users\GOAT\humayra\buet\3-1\eee 312\project final\features";
if ~exist(features_dir, 'dir')
mkdir(features_dir);
end
windowLength = round(0.03 * Fs); % 30ms window
overlapLength = round(0.025 * Fs); % 25ms overlap
win = hann(windowLength, 'periodic');
aFE = audioFeatureExtractor(...
'SampleRate', Fs, ...
'Window', win, ...
'OverlapLength', overlapLength, ...
'mfcc', true, ...
'mfccDelta', true, ...
'mfccDeltaDelta', true, ...
'pitch', true, ...
'spectralCentroid', true, ...
'zerocrossrate', true, ...
'shortTimeEnergy', true);
files = dir(fullfile(standardized_dir, '**', '*.wav'));
numMFCC = 13;
featureNames = ["Label"];
for i = 1:numMFCC
featureNames = [featureNames, "MFCC" + i, "MFCCDelta" + i, "MFCCDeltaDelta" + i];
end
```

```
featureNames = [featureNames, "Pitch", "SpectralCentroid", "ZeroCrossRate",
"ShortTimeEnergy"];
featureTable = table('Size', [0, length(featureNames)], 'VariableTypes', repmat("double",
1, length(featureNames)), 'VariableNames', featureNames);
for i = 1:length(files)
file = files(i);
audioFilePath = fullfile(file.folder, file.name);
[y, ~] = audioread(audioFilePath);
features = extract(aFE, y);
newRow = zeros(1, length(featureNames));
tokens = regexp(file.name, '(\d+)_\d+\.wav', 'tokens');
if ~isempty(tokens)
label = str2double(tokens{1}{1});
else
continue;
end
newRow(1) = label;
featureIndex = 2;
for j = 1:numMFCC
newRow(featureIndex) = mean(features(:, j)); % MFCC
newRow(featureIndex + 1) = mean(features(:, numMFCC + j)); % MFCCDelta
newRow(featureIndex + 2) = mean(features(:, 2*numMFCC + j)); % MFCCDeltaDelta
featureIndex = featureIndex + 3;
end
newRow(featureIndex) = mean(features(:, end-3)); % Pitch
newRow(featureIndex + 1) = mean(features(:, end-2)); % SpectralCentroid
newRow(featureIndex + 2) = mean(features(:, end-1)); % ZeroCrossRate
newRow(featureIndex + 3) = mean(features(:, end)); % ShortTimeEnergy
featureTable = [featureTable; array2table(newRow, 'VariableNames', featureNames)];
end
disp('Saving features...');
save(fullfile(features_dir, 'extracted_features_table.mat'), 'featureTable');
disp('Features saved.');
disp('Feature extraction and saving completed.');
%% Label vs Feature Scatter Plots
featureNames = featureTable.Properties.VariableNames;
% numFeatures = width(featureTable);
numFeatures = 3;
% Loop through each feature to create a scatter plot
for i = 2:numFeatures
figure;
scatter(featureTable.(featureNames{i}), featureTable.Label);
ylabel('Label');
xlabel(featureNames{i});
title(featureNames{i});
end
%% Training and Testing
labels = featureTable{:, 1};
features = featureTable{:, 2:end};
M = mean(features, 1);
S = std(features, [], 1);
features = (features - M) ./ S;
k = 3;
predictedLabels = zeros(1, size(features, 2));
save('trained_model.mat', 'M','S','features','labels','k');
%% RealTime Testing
elseif choice == 3
load('trained_model.mat');
Fs = 16000; % Sampling frequency
% Record user's voice
disp('Press any key to start recording');
pause;
```

```
disp('Speak Now!');
recobj = audiorecorder(Fs, 16, 1);
recordblocking(recobj, 3);
y = getaudiodata(recobj);
disp('Recording completed.');
% Define the feature extraction parameters
windowLength = round(0.03 * Fs); % 30ms window
overlapLength = round(0.025 * Fs); % 25ms overlap
win = hann(windowLength, 'periodic');
aFE = audioFeatureExtractor(...
'SampleRate', Fs, ...
'Window', win, ...
'OverlapLength', overlapLength, ...
'mfcc', true, ...
'mfccDelta', true, ...
'mfccDeltaDelta', true, ...
'pitch', true, ...
'spectralCentroid', true, ...
'zerocrossrate', true, ...
'shortTimeEnergy', true);
% Apply preprocessing steps to the captured audio
Fc = 3000; % Cut-off frequency for the low pass filter
order = 5; % Order of the filter
y_filtered = designfilt('lowpassfir', 'FilterOrder', order, 'CutoffFrequency', Fc,
'SampleRate', Fs);
y_filtered2 = filtfilt(y_filtered, y);
y_filtered = removing_silentpart(y_filtered2, Fs);
y_normalized = (y_filtered - mean(y_filtered)) / std(y_filtered);
% Extract features from the preprocessed real-time audio data
realTimeFeatures = [];
if ~any(isnan(y_normalized))
realTimeFeatures = extract(aFE, y_normalized);
else
disp('Error: Input signal contains NaN values.');
end
realTimeFeatures = (realTimeFeatures - M) ./ S;
% Predict label for each feature
for featureIndex = 1:size(features, 2)
predictedLabels(featureIndex) = custom_knn(features(:, featureIndex), labels,
realTimeFeatures(featureIndex), k);
end
% Final predicted label by majority voting across all features
finalPredictedLabel = mode(predictedLabels);
% Count how many predicted labels match the final predicted label
matchingCount = sum(predictedLabels == finalPredictedLabel);
% Calculate the matching percentage
totalFeatures = size(features, 2); % Total number of features used for prediction
matchingPercentage = (matchingCount / totalFeatures) * 100;
if matchingPercentage >= 23
disp(['Predicted Student Id: ', num2str(finalPredictedLabel)]);
fprintf('Matching Percentage: %.2f%%\n', matchingPercentage);
else
disp("Proxy Attendance is not allowed");
end
%Visualisation
featureIndex = 1; % For example, 1 for MFCC1
featureData = features(:, featureIndex);
realTimeFeature = realTimeFeatures(featureIndex);
% Calculate distances and find nearest neighbors as before
distances = abs(featureData - realTimeFeature);
[sortedDistances, sortedIndices] = sort(distances);
kNearestIndices = sortedIndices(1:k);
```

```
% Scatter plot for the feature vs. labels
figure;
scatter(featureData, labels, 'b', 'o'); % Blue circles for existing data
hold on;
scatter(realTimeFeature, finalPredictedLabel, 'r', 'filled'); % Red filled circle for
real-time feature
xlabel(['Feature ' num2str(featureIndex)]);
ylabel('Label');
title(['Scatter Plot for Feature ' num2str(featureIndex) ' vs. Label']);
% Draw lines to the 3 nearest neighbors
for i = 1:k
line([realTimeFeature, featureData(kNearestIndices(i))], [finalPredictedLabel,
labels(kNearestIndices(i))], 'Color', 'k', 'LineStyle', '--');
end
hold off;
legend({'Existing Data', 'Real-Time Feature'}, 'Location', 'best');
save('finalPredictedLabel.mat', 'finalPredictedLabel');
%% Attendance Database
elseif choice ==4
% Load or Initialize the Attendance Database
load('finalPredictedLabel.mat');
if exist('attendance_database.mat', 'file')
load('attendance_database.mat', 'attendanceTable');
else
attendanceTable = table([], [], 'VariableNames', {'StudentID', 'AttendanceRecord'});
end
% Check if the student ID is already in the database
studentID = num2str(finalPredictedLabel);
studentIndex = find(strcmp(attendanceTable.StudentID, studentID));
if isempty(studentIndex)
newEntry = {studentID, 'P'};
attendanceTable = [attendanceTable; cell2table(newEntry, 'VariableNames', {'StudentID',
'AttendanceRecord'})];
else
attendanceTable.AttendanceRecord{studentIndex} =
strcat(attendanceTable.AttendanceRecord{studentIndex}, 'P');
end
% Save the updated attendance database
save('attendance_database.mat', 'attendanceTable');
% Prompt the user for the total number of working days
totalWorkingDays = input('Enter the total number of working days: ');
% Calculate and display attendance summaries
calculating_attendance_summary(attendanceTable, totalWorkingDays);
end
end
```

%% Functions

```
function [new_signal]=removing_silentpart(y,Fs)
% removing silent parts
frame_duration=0.01; % Frame length is 10ms
frame_length=Fs*frame_duration;
N=length(y);
frame_number=floor(N/frame_length);
new_signal=zeros(N,1);
count=1;
threshold=0.05;
for i=1:frame_number
% Extracting the frame
frame=y((((i-1)*frame_length)+1):(frame_length*i));
% Finding the maximum value of the frame
```

```
max_val=max(frame);
% Finding the silent frame and removing it
if (max_val>threshold)
count=count+1;
new_signal((((count-1)*frame_length)+1):(frame_length*count))=frame;
end
end
% End point detection
post=find(new_signal~=0);
length1=length(new_signal);
new_signal(post(end):length1)=[];
end

%Recording audio
fs = 16000;
rec_obj = audiorecorder(fs,16,1);
rec_duration = 3;
disp('Start speaking.');
recordblocking(rec_obj,rec_duration);
disp('End of recording.');
my_recording = getaudiodata(rec_obj);
filename = 'my_recording.wav';
audiowrite(filename, my_recording, fs);
figure(1);
plot(my_recording);
signal = my_recording;
%Window parameters
frame_size = 0.020; %frame size is 20ms
frame_stride = 0.01; %frame overlap is 10ms
frame_length = frame_size * fs; %frame size in samples
frame_step = frame_stride * fs; %frame step in samples
signal_length = length(signal);
frame_number = ((signal_length-frame_length)/frame_step);
w = hann(frame_length);
n = 1:frame_length;
figure(2);
plot(n,w);
X = zeros(512,frame_number);
%Fourier Transform and Spectrogram
for k = 1:frame_number
start_index = (k-1) * frame_step + 1;
end_index = start_index + frame_length - 1;
frame = signal(start_index:end_index);
windowed_frame = frame .* w;
frame_fft = fft(windowed_frame, 512);
X(:, k) = frame_fft;
end
figure(3);
spectrogram(signal, frame_length, frame_length-frame_step, 512, fs, 'yaxis');
title('Spectrogram of the signal');
disp(size(X));
%Power Spectral Density Estimation
P = (1 / frame_length) * abs(X).^2;
P = P(1:size(P,1)/2+1, :);
figure(4);
imagesc(20*log10(P));
axis xy;
xlabel('Frame Number');
ylabel('Frequency Bin');
title('Periodogram of the signal');
colorbar;
frame_to_plot = 1;
```

```
frequencies = linspace(0, fs/2, size(P,1));
figure(5);
plot(frequencies, 20*log10(P(:, frame_to_plot)));
xlabel('Frequency (Hz)');
ylabel('Power/Frequency (dB/Hz)');
title(['Periodogram of Frame ' num2str(frame_to_plot)]);
grid on;
%Mel Filterbank Processing
lower_freq = 300;
upper_freq = 8000;
lower_mel = freq_to_mel(lower_freq);
upper_mel = freq_to_mel(upper_freq);
num_filters = 26;
mels = linspace(lower_mel, upper_mel, num_filters + 2);
hz = mel_to_freq(mels);
f_bin = floor((257 + 1) * hz / (fs/2));
filterbank = zeros(num_filters, 257);
for m = 1:num_filters
for k = 1:512
if f_bin(m) <= k && k <= f_bin(m+1)
filterbank(m, k) = (k - f_bin(m)) / (f_bin(m+1) - f_bin(m));
elseif f_bin(m+1) < k && k <= f_bin(m+2)
filterbank(m, k) = (f_bin(m+2) - k) / (f_bin(m+2) - f_bin(m+1));
end
end
end
figure(6);
hold on;
for n = 1:num_filters
plot((0:257-1) * ((fs/2) / 257), filterbank(n, :));
end
xlabel('Frequency (Hz)');
ylabel('Amplitude');
title('The 26-filter Mel Filterbank');
%MFCC coefficient calculation
filtered_energies = filterbank * P;
log_filterbank_energies = log(filtered_energies + eps);
my_mfcc_full = dct(log_filterbank_energies);
my_mfcc = my_mfcc_full(1:13, :);
% Plotting the Mel Spectrogram
figure(7);
imagesc(20 * log10(filtered_energies + eps)); % Convert energies to dB
axis xy;
xlabel('Frame Number');
ylabel('Mel Filter Number');
title('Mel Spectrogram');
colorbar;
% Plotting the MFCC Coefficients
figure(8);
imagesc(my_mfcc);
axis xy;
xlabel('Frame Number');
ylabel('MFCC Coefficients');
title('MFCC Coefficients');
colorbar;
%Delta coefficent calculation
delta_window = 2; % N = 2
padding = repmat(my_mfcc(:,1), 1, delta_window);
padded_mfcc = [padding, my_mfcc, padding];
deltas = zeros(size(my_mfcc));
for t = 1 + delta_window : size(padded_mfcc, 2) - delta_window
numerator = zeros(size(my_mfcc, 1), 1);
```

```
denominator = 0;
for n = 1:delta_window
numerator = numerator + n * (padded_mfcc(:, t + n) - padded_mfcc(:, t - n));
denominator = denominator + 2 * n^2;
end
deltas(:, t - delta_window) = numerator / denominator;
end
padding = repmat(deltas(:,1), 1, delta_window);
padded_deltas = [padding, deltas, padding];
%Delta-delta coefficient calculation
delta_deltas = zeros(size(deltas));
for t = 1 + delta_window : size(padded_deltas, 2) - delta_window
numerator = zeros(size(deltas, 1), 1);
denominator = 0;
for n = 1:delta_window
numerator = numerator + n * (padded_deltas(:, t + n) - padded_deltas(:, t - n));
denominator = denominator + 2 * n^2;
end
delta_deltas(:, t - delta_window) = numerator / denominator;
end
combined_mfcc = [my_mfcc; deltas; delta_deltas];
%Comparison
[audioMFCC,~,~] = mfcc(signal, fs, 'LogEnergy', 'Ignore');
% Compare MFCCs
corrMFCC = corrcoef(my_mfcc(:), audioMFCC(:));
mseMFCC = immse(my_mfcc(:), audioMFCC(:));
% Display results
disp(['Correlation between custom and built-in MFCCs: ', num2str(corrMFCC(1,2))]);
disp(['MSE between custom and built-in MFCCs: ', num2str(mseMFCC)]);
function mel = freq_to_mel(freq)
mel = 1125 * log(1 + freq / 700);
end
function freq = mel_to_freq(mel)
freq = 700 * (exp(mel / 1125) - 1);
end

function predictedLabel = custom_knn(featureData, labels, realTimeFeature, k)
distances = abs(featureData - realTimeFeature);
[~, sortedIndices] = sort(distances);
kNearestIndices = sortedIndices(1:k);
kNearestLabels = labels(kNearestIndices);
predictedLabel = mode(kNearestLabels);
end

function calculating_attendance_summary(attendanceTable, totalWorkingDays)
numStudents = height(attendanceTable);
studentIDs = attendanceTable.StudentID;
attendancePercentages = zeros(numStudents, 1);
for i = 1:numStudents
totalDays = totalWorkingDays;
presentDays = count(attendanceTable.AttendanceRecord{i}, 'P');
attendancePercentage = (presentDays / totalWorkingDays) * 100;
attendancePercentages(i) = attendancePercentage;
fprintf('Student ID: %s, Total Days: %d, Present Days: %d, Attendance
Percentage: %.2f%%\n', ...
studentIDs{i}, totalDays, presentDays, attendancePercentage);
end
% Plotting the attendance percentages
figure;
bar(attendancePercentages);
title('Attendance Percentage of Each Student');
```

```
xlabel('Student Index');
ylabel('Attendance Percentage (%)');
xticks(1:numStudents);
xticklabels(studentIDs);
xtickangle(45);
ylim([0 100]);
grid on;
end
```

Table: Source Code for the main program

4 Implementation

4.1 Description

The implementation model consists of a four-step procedure:

1. Creating a central database: Using an application, each student must register their voice. The voice samples that will be utilized for program training and cross-checking student identity will be stored in this central database. Additionally, it will provide each student's automatically updated attendance information.
2. Developing an app for the system of attendance: There will be an app made to record attendance. There will be four options on the app interface: for recording voice notes, and for entering attendance, for training the model and lastly for viewing the attendance database.
3. Giving each faculty member a gadget: At the start of class, each faculty member will have a device that they can distribute across the classroom. Students will utilize the gadget, which might be a tablet with a dedicated, high-quality noise-cancelling microphone, to record their attendance. The students will use the program to record their attendance by speaking their student ID into the microphone. After recording their attendance, they will pass it on to the following individual.
4. Additional database updates: Since the institution admits new students every year, the voice samples of all first-year students are gathered centrally by the university administration before to the start of classes by setting up a separate booth. After that, starting with the first week of courses, the students can easily transition to the biometric attendance system.

5 Design Analysis and Evaluation

5.1 Novelty

Our project introduces a novel approach to classroom management through the implementation of biometric authentication using voice recognition technology. Unlike traditional attendance tracking methods, our system offers real-time testing and authentication capabilities, enabling instant recording of attendance as students speak. This is facilitated by customized preprocessing techniques, such as filtering and removal of silent parts, which ensure accurate voice recognition even in noisy classroom environments. Leveraging machine learning algorithms for classification, our system continuously learns from new data, improving its accuracy over time and adapting to changes in student voice patterns. Seamless integration into existing workflows and a user-friendly interface promote widespread adoption and acceptance among educators and students. Moreover, our project prioritizes ethical considerations and privacy protection by implementing secure data storage and access control mechanisms, ensuring compliance with relevant regulations and guidelines. In summary, our project represents a cutting-edge solution that enhances efficiency, security, and engagement in educational settings through innovative voice-based authentication for

classroom management.

5.2 Design Considerations (PO(c))

5.2.1 Considerations to public health and safety

In designing our biometric authentication system for classroom management, public health and safety considerations are paramount. We ensure that the system's implementation, including the use of voice recognition technology and data handling processes, complies with relevant health and safety regulations. This involves minimizing physical contact points by utilizing voice-based authentication methods, reducing the risk of transmission of contagious diseases within educational environments. Additionally, measures are implemented to protect the privacy and security of biometric data, mitigating potential risks associated with unauthorized access or misuse of sensitive information.

5.2.2 Considerations to environment

Environmental considerations are considered throughout the design and implementation of our system. We prioritize the use of energy-efficient hardware components and strive to minimize power consumption during system operation. Additionally, efforts are made to reduce electronic waste by designing the system for longevity and ease of upgradeability, thereby extending its lifespan and reducing the need for frequent replacements. Furthermore, where applicable, materials and manufacturing processes are chosen with sustainability in mind, aiming to minimize the system's overall environmental footprint.

5.2.3 Considerations to cultural and societal needs

In addressing cultural and societal needs, our design approach emphasizes inclusivity and accessibility. We ensure that the authentication system is designed to accommodate diverse cultural backgrounds and linguistic preferences, allowing for flexible enrollment and authentication processes that are sensitive to individual identities and preferences. Additionally, efforts are made to promote user education and awareness regarding the system's operation and implications, fostering transparency and trust within the educational community. Moreover, we prioritize the integration of features that enhance accessibility for individuals with disabilities, ensuring equitable access to educational resources and opportunities for all students. Through these considerations, our system aims to support and enrich cultural diversity while fostering a safe and inclusive learning environment.

5.3 Investigations (PO(d))

In our endeavor to develop a biometric authentication system for classroom management, a series of investigations are undertaken to ensure its efficacy, reliability, and ethical integrity. These investigations span several critical domains: technical feasibility, wherein the compatibility and robustness of voice recognition technology within educational settings are assessed; user acceptance and usability, where feedback from educators, students, and administrative staff informs iterative refinements to the system's design; security and privacy, with rigorous evaluations conducted to identify and mitigate potential vulnerabilities while safeguarding biometric data; and ethical considerations, guiding decision-making to ensure transparency, fairness, and respect for individual autonomy and privacy. Through these investigations, our goal is to create a biometric authentication system that not only meets technical requirements but also addresses user needs, upholds ethical principles, and

contributes to a safe, inclusive, and conducive learning environment.

5.3.1 Design of Experiment

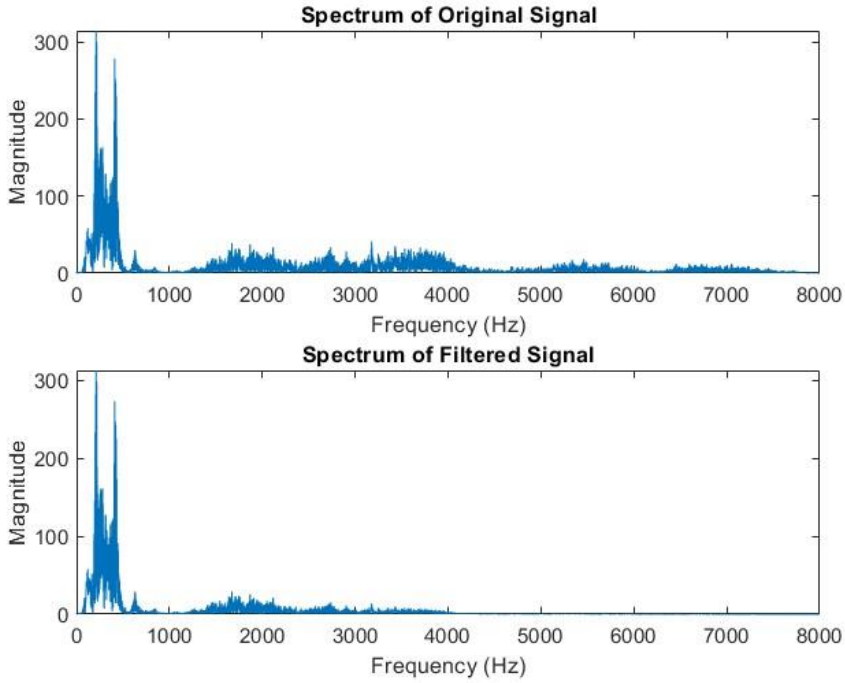
In our experimental approach, we initiated by implementing the biometric authentication system, followed by the systematic collection of data from students. Subsequently, a series of tests were conducted to evaluate the system's capability to accurately recognize speakers. These tests encompassed diverse scenarios and conditions to assess the system's robustness and performance under varying circumstances. Additionally, we incorporated a graphical interface system to enhance user interaction and facilitate seamless operation of the authentication system. The graphical interface not only provides a user-friendly experience but also enables convenient monitoring and management of authentication processes. By integrating these elements into our experimental design, we aimed to comprehensively evaluate the effectiveness, reliability, and usability of our biometric authentication system, thereby informing iterative improvements and optimizations to enhance its functionality and user experience.

5.3.1.1 Data Collection

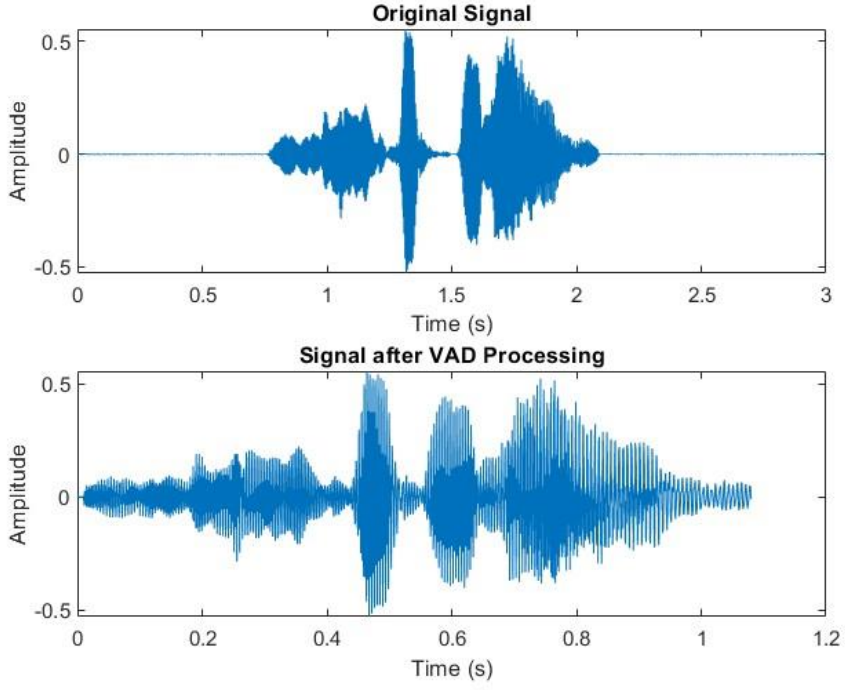
In our project, we aimed to streamline the collection and organization of voice data for our voice-based attendance system by implementing a structured process. By prompting users to input their student ID, we created individual directories for each participant, ensuring organized data storage. Our recordings, set at a 16000Hz sampling frequency for optimal voice recognition, were designed to be both high-quality and user-friendly. Each 3-second audio snippet was automatically saved using the student's ID and a sequence number, facilitating efficient data management and laying a solid foundation for the accurate recognition of each student's unique voice characteristics in the system.

5.3.1.2 Pre-processing

In the development of our voice-based attendance system, a meticulous process was undertaken to preprocess the collected voice data, aiming to optimize its quality for subsequent analysis and recognition tasks. The initial phase of this process involved filtering raw audio signals to mitigate the impact of high-frequency noise and other non-essential sounds that could potentially obscure the characteristics vital for accurate voice recognition. Utilizing a fifth order low-pass filter with a cut-off frequency of 3000Hz, our objective was to isolate the core frequencies that predominantly convey speech information, thereby improving the quality of the voice data.

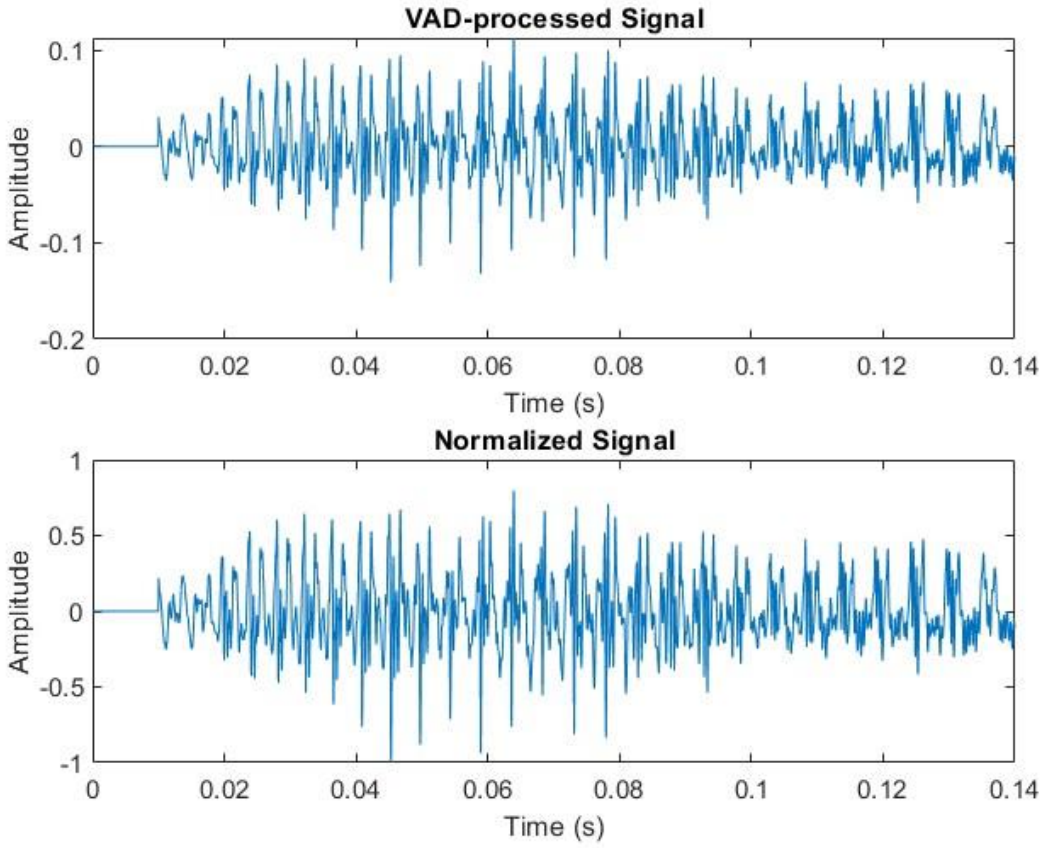


A significant focus was placed on Voice Activity Detection (VAD), where a custom function played a pivotal role in distinguishing between speech and non-speech segments within the audio samples. This custom function meticulously analyzed each frame of the audio, employing a frame duration of 0.01 seconds to closely monitor the signal's amplitude variations. By setting a threshold of 0.05 for amplitude levels, the function effectively identified and retained frames that contained voice activity, thereby excluding silent or irrelevant parts. This selective process was crucial in reducing the dataset to its most essential elements, minimizing computational demands and enhancing the efficiency of the system.



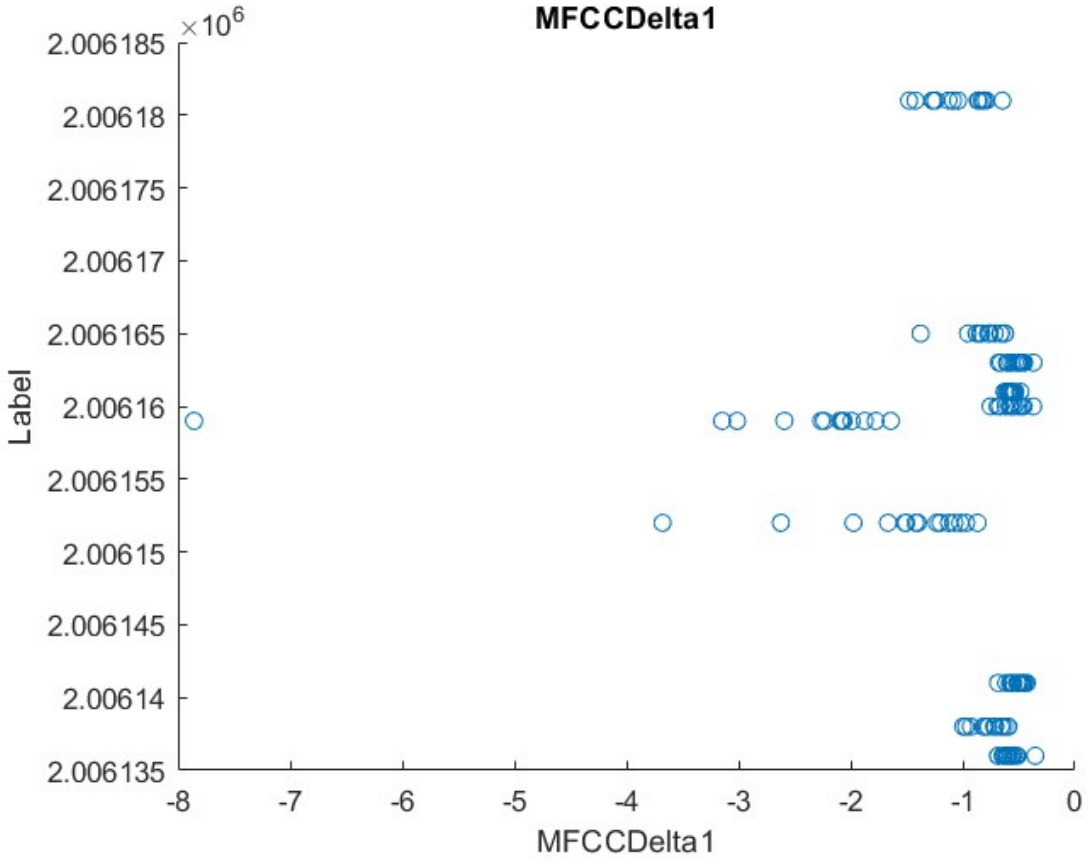
In the final stage of data preprocessing, we implemented standardization and normalization across the voice samples. This involved adjusting each audio sample to achieve a zero mean and scaling them to maintain a uniform standard deviation. These steps were critical for normalizing the data distribution, ensuring that variations in volume and intensity across

different recordings did not affect the model's ability to accurately identify and verify student identities.



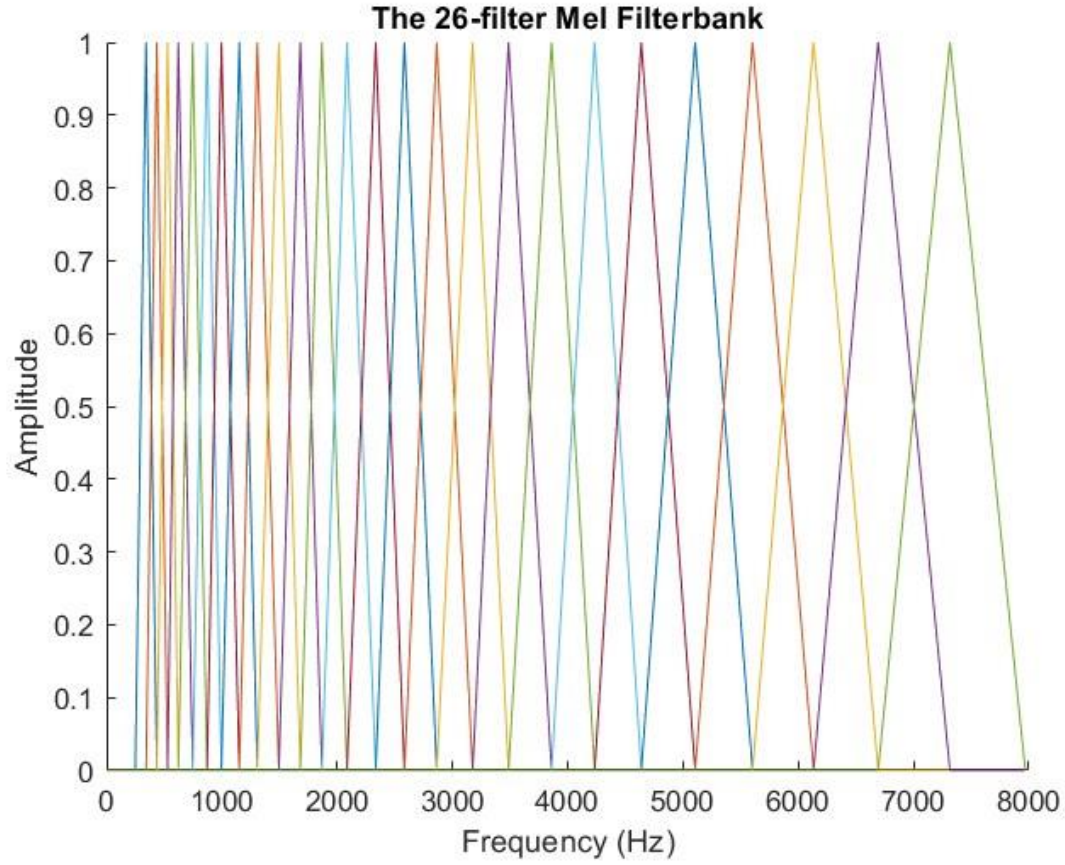
5.3.1.3 Feature Extraction

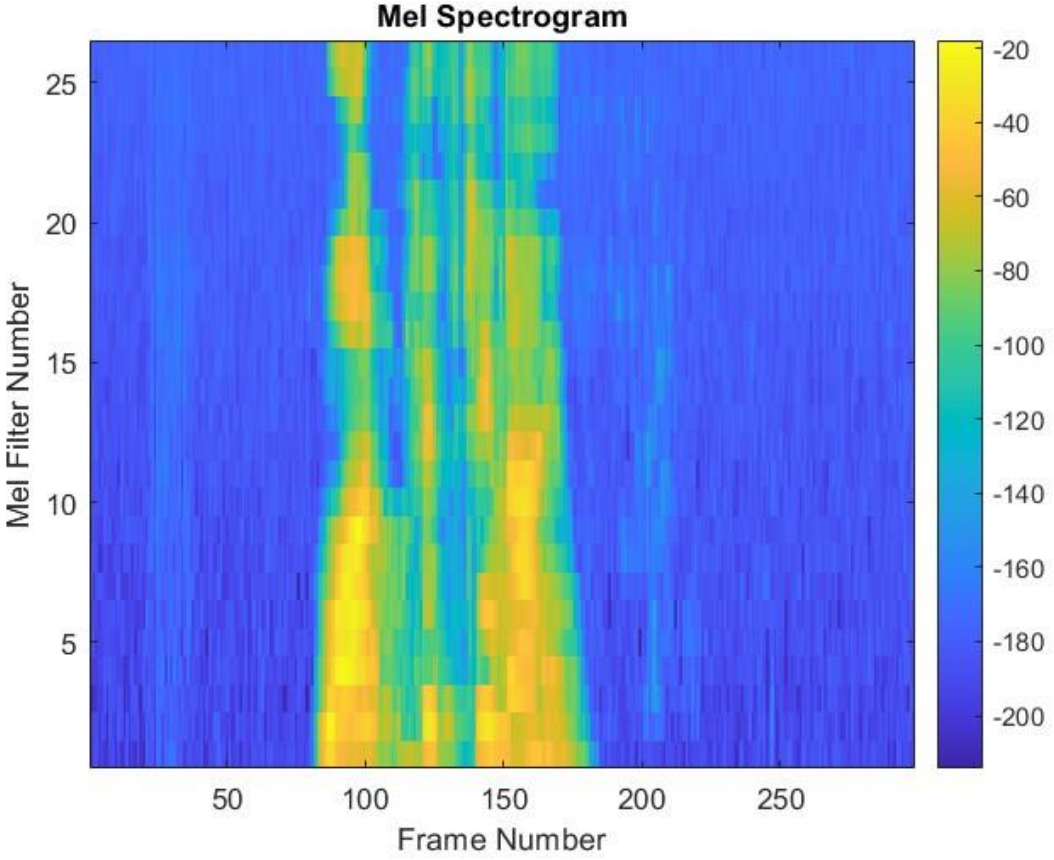
In the feature extraction phase of our voice-based attendance system, our primary goal was to accurately capture and quantify the unique vocal characteristics of each student to facilitate reliable identification. To achieve this, we utilized a comprehensive suite of audio features, including Mel-frequency cepstral coefficients (MFCCs) and their deltas, pitch, spectral centroid, zero-cross rate, and short-time energy. These features were extracted using a window length of 30ms and overlap length of 25ms to ensure the capture of both temporal and spectral information crucial for voice recognition. The choice of features like MFCCs and their deltas was driven by their effectiveness in representing the vocal tract configuration, which is unique to everyone, while pitch and spectral features capture the tone and quality of the voice. This methodical approach allowed us to build a robust feature set conducive to accurate speaker identification.

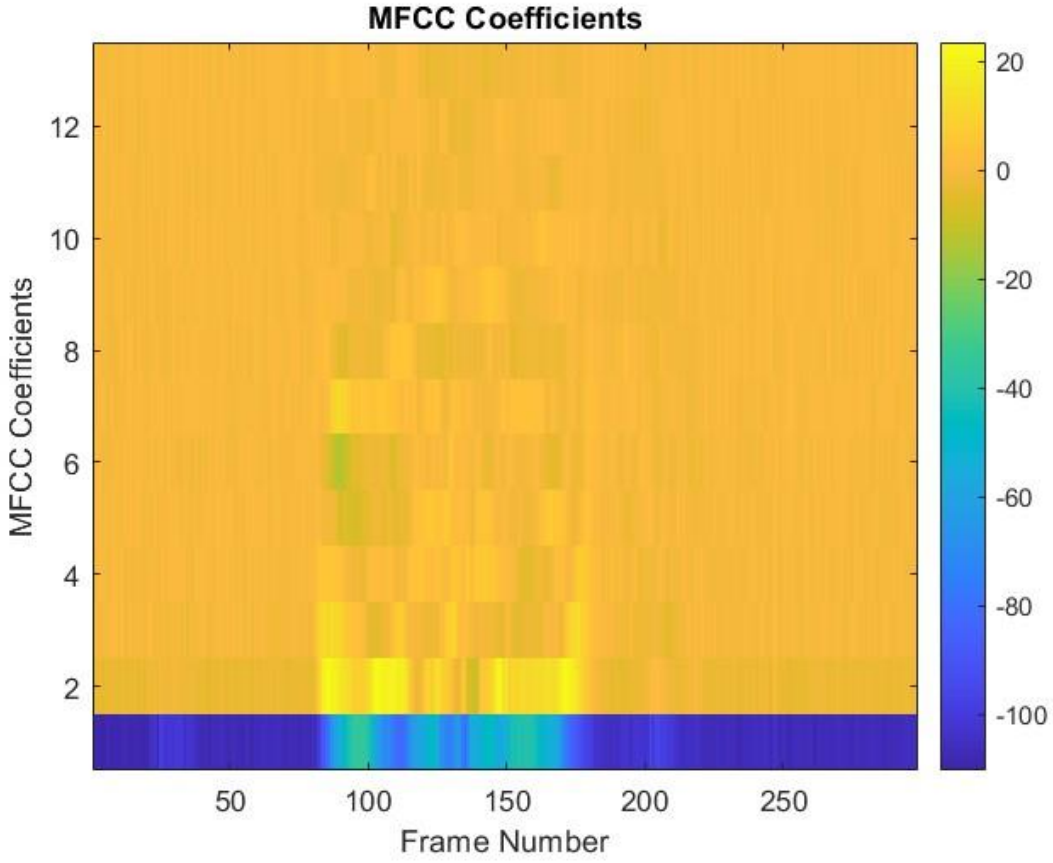


1 Label	2 MFCC1	3 MFCCDelta1	4 MFCCDeltaDelta1	5 MFCC2	6 MFCCDelta2	7 MFCCDeltaDelta2	8 MFCC3	9 MFCCDelta3	10 MFCCDeltaDelta3	11 MFCC4	12 MFCCDelta4	13 MFCCDeltaDelta4	14 MFCC5	15 MFCCDelta5	16 MFCCDeltaDelta5	17 MFCC6	18 MFCCDelta6	19 MFCCDeltaDelta6	20 MFCC7	21 MFCCDelta7
2006136	-39.1222	-0.5655	-0.0360	10.7120	0.1481	0.0136	-0.7987	-0.0183	-1.6435e-04	1.3881	0.0373	0.0013	0.5319	0.0428	0.0075	0.1577	0.0094	0.0037	-0.2232	-0.0167
2006136	-38.2916	-0.6355	-7.5511e-04	7.9312	0.0106	-0.0095	-2.9083	-0.1024	0.0051	2.8458	0.0990	0.0014	0.1589	-3.9657e-04	-2.8580e-04	-0.4083	-0.0440	-0.0085	0.7597	0.0218
2006136	-38.4127	-0.6164	0.0071	9.2420	0.0016	-0.0273	-2.6073	-0.0945	0.0089	2.0515	0.0387	-0.0132	0.3317	-0.0032	3.8591e-04	0.0897	0.0057	0.0014	0.9304	0.0565
2006136	-40.5378	-0.5480	-0.0014	8.5418	0.0271	-0.0014	-2.8874	-0.1374	-1.8645e-04	2.5155	0.0546	-5.0850e-04	0.2068	-0.0045	-4.6035e-04	-0.1596	-0.0154	9.3504e-04	0.8496	0.0451
2006136	-40.9879	-0.5271	-0.0370	9.1283	0.1591	0.0109	-1.4588	0.0351	0.0068	2.7238	0.0252	0.0035	-0.0788	0.0110	0.0031	-0.6822	-0.0091	-0.0023	0.3257	-0.0046
2006136	-40.4051	-0.6104	1.8417e-04	8.2519	0.0279	0.0017	-2.4399	-0.1013	0.0031	2.8444	0.0910	0.0018	-0.0555	-0.0361	-0.0024	-0.6488	-0.0565	-0.0037	0.6569	0.0417
2006136	-37.7237	-0.6444	-0.0101	8.8797	-0.0075	-0.0297	-2.1446	-0.0714	-0.0123	2.2128	0.0678	0.0015	0.7614	0.0017	-0.0092	-0.2362	-0.0280	-0.0051	-0.0627	0.0307
2006136	-41.8559	-0.5133	0.0027	10.2458	0.0145	-0.0044	-2.0084	-0.0959	0.0079	1.8211	0.0560	-8.2655e-04	0.1472	-0.0093	-5.6232e-04	0.1646	-0.0097	-0.0032	0.7003	0.0443
2006136	-39.8389	-0.6822	0.0122	10.7651	0.0886	-0.0305	-0.4432	0.0259	4.0025e-04	1.0941	0.0726	0.0071	0.3549	0.0117	-0.0106	0.2249	-9.3998e-04	-0.0038	-0.0098	0.0274
2006136	-38.6717	-0.5933	0.0077	9.5124	0.0050	-0.0073	-2.7614	-0.1199	0.0115	2.0385	0.0349	-0.0055	0.2383	0.0024	0.0023	-0.1376	-0.0043	-9.0005e-04	0.6772	0.0385
2006136	-39.4153	-0.5943	-0.0215	10.6518	0.1552	0.0029	-0.5947	-0.0155	2.4027e-04	1.2169	0.0484	0.0057	0.2080	0.0475	0.0076	0.0081	5.3222e-04	9.5039e-04	-0.1153	-0.0265
2006136	-40.2397	-0.6865	-0.0062	8.9789	0.0616	5.4657e-04	-2.8490	-0.0797	0.0101	3.6190	0.1396	0.0044	0.1074	-0.0075	-0.0049	-0.3470	-0.0487	-0.0091	1.3434	0.0495
2006136	-40.2533	-0.6040	-3.3749e-04	8.6330	0.0185	-0.0019	-3.4616	-0.1213	-4.7439e-04	3.0090	0.0770	-8.2342e-04	0.4699	0.0019	0.0017	-0.3673	-0.0308	9.8223e-05	1.0903	0.0295
2006136	-40.5624	-0.5343	0.0114	9.4009	0.0787	-0.0105	-2.8122	0.0203	0.0070	3.0698	0.0138	-0.0049	0.9085	0.0077	-0.0089	-0.1878	0.0013	0.0033	0.6078	-0.0015
2006136	-40.2144	-0.5734	0.0017	10.1302	0.1296	0.0014	-1.7362	-0.0698	-0.0032	2.8259	0.1099	-0.0044	0.5418	0.0210	3.7184e-04	-0.2259	-0.0095	0.0021	0.2000	0.0252
2006136	-38.5303	-0.5994	-0.0167	9.7620	0.1818	0.0128	-1.3990	0.0179	-0.0036	0.7724	0.0097	0.0022	0.3199	0.0047	0.0030	-0.3055	-0.0050	-0.0013	-0.7382	-0.0067
2006138	-40.8891	-0.7220	0.0212	9.5841	0.1432	-0.0180	-1.0445	0.0398	0.0013	2.1442	0.0433	-0.0047	0.6556	0.0190	-2.2327e-04	-0.6956	0.0043	0.0046	-0.5582	0.0066
2006138	-39.2407	-0.8034	-0.0406	10.2135	0.2334	0.0162	-1.5010	-6.6796e-04	0.0016	1.6963	0.0551	0.0045	0.8878	0.0674	0.0111	-0.5752	3.2523e-05	-1.8823e-04	-0.8140	-0.0315
2006138	-38.3033	-0.7102	-0.0154	9.7402	0.1606	-0.0049	-2.7437	-0.1438	-0.0292	1.1920	0.0977	0.0120	0.2975	0.0478	0.0019	-0.3406	-0.0036	-0.0012	-0.3919	0.0216
2006138	-42.5835	-0.8167	-0.0077	10.6753	0.1517	0.0106	-1.1901	-0.1586	-0.0161	1.4117	0.0834	-0.0074	0.6696	0.0587	-0.0017	-0.2744	0.0117	0.0048	-0.4030	0.0335
2006138	-40.8919	-0.9973	-0.0115	9.8893	0.1977	0.0036	-1.7158	-0.2045	-0.0540	2.3247	0.1652	0.0262	0.5409	0.0726	0.0038	-1.0104	-0.0233	-0.0062	-0.4771	0.0282
2006138	-40.9551	-0.9722	-0.0461	10.6648	0.2109	-0.0024	0.1639	0.0423	0.0169	0.8256	0.0414	0.0034	0.2409	0.0308	0.0039	-0.1211	0.0110	0.0038	-0.7884	-0.0227
2006138	-39.1373	-0.6331	-0.0085	9.7389	0.0788	-0.0039	-2.6400	-0.1601	-7.6199e-04	1.3324	0.0264	-0.0064	0.7681	0.0249	-0.0015	-0.2388	0.0157	0.0016	-0.4315	0.0396
2006138	-40.0056	-0.6571	-0.0110	10.0843	0.0893	-1.0326e-04	-2.7786	-0.1837	-0.0105	1.3110	0.0085	-0.0057	0.8641	-3.1977e-04	-0.0053	-0.4196	-0.0267	-0.0031	-0.7778	0.0136
2006138	-38.6935	-0.7720	-0.0160	9.7434	0.0852	-0.0242	-2.0924	-0.1908	-0.0487	1.1573	0.0436	-0.0047	0.4904	0.0265	-0.0089	-0.3901	-0.0139	-0.0057	-0.6982	0.0187
2006138	-40.3188	-0.6005	-0.0138	10.3242	0.1084	0.0019	-0.5763	0.0366	0.0031	1.6394	0.0311	7.0211e-04	1.1433	0.0193	0.0011	0.0644	0.0119	7.1388e-04	-0.8789	0.0058
2006138	-39.3130	-0.5937	0.0478	11.1661	0.1470	-0.0155	-1.3962	-0.0925	-0.0393	1.3840	0.0407	-0.0041	0.3900	0.0206	-0.0018	-0.7552	-0.0233	-0.0018	-0.5729	0.0025
2006138	-41.9194	-0.9279	-0.0321	10.9234	0.2356	-3.7198e-04	-1.3049	0.0619	0.0060	1.1282	0.0739	0.0088	0.5876	0.0357	0.0027	-0.6520	-0.0027	-7.0479e-04	-0.7692	0.0080
2006138	-39.8630	-0.6605	-7.8125e-04	10.2227	0.1348	1.9950e-04	-2.0134	-0.1378	-0.0284	1.8912	0.0803	0.0092	0.8583	0.0474	-0.0016	-0.6248	-0.0075	-0.0025	-0.5445	0.0186
2006138	-39.7316	-0.7984	-0.0251	10.5499	0.2376	0.0088	-1.0233	-0.0105	-2.3526e-04	0.7796	0.0536	0.0043	0.4391	0.0583	0.0076	-0.4675	-0.0102	-0.0019	-1.0242	-0.0428
2006141	-35.9584	-0.6134	-0.0159	9.9677	0.1863	0.0094	-0.7547	-0.0011	0.0056	1.1033	0.0221	-5.0405e-05	0.3458	0.0147	-0.0023	0.0547	0.0079	-0.0011	-0.7708	-0.0135
2006141	-36.7582	-0.5788	-0.0200	11.1043	0.1796	0.0070	-0.0102	-0.0050	0.0040	0.9655	0.0247	0.0038	0.0173	0.0146	0.0029	-0.1414	0.0101	0.0041	-0.3130	-0.0064
2006141	-37.4777	-0.4625	0.0036	11.1950	0.1368	-0.0022	0.1085	-0.0177	8.0073e-04	0.5779	0.0021	-0.0020	-0.4732	-0.0056	7.8076e-04	-0.2044	-0.0048	0.0015	-0.3925	-0.0054
2006141	-35.3992	-0.5003	-0.0094	10.2507	0.1340	-0.0028	-0.7952	0.0038	0.0052	1.0079	0.0081	0.0012	0.2353	0.0129	0.0044	-0.3637	0.0054	0.0034	-0.6970	-0.0038
2006141	-35.9092	-0.4658	-0.0146	10.5155	0.1501	0.0074	-0.5627	-0.0208	-5.8859e-04	0.9906	0.0105	8.0769e-04	0.0812	0.0046	0.0016	-0.3030	7.4131e-05	0.0019	-0.5562	-0.0042
2006141	-35.5909	-0.4306	-0.0070	10.1094	0.1293	0.0042	-0.7389	-0.0176	0.0012	0.9646	-0.0020	-5.5736e-04	0.1399	-0.0037	0.0016	-0.3354	-7.1180e-04	0.0019	-0.7035	-0.0015
2006141	-35.0630	-0.5482	-0.0325	10.0930	0.1936	0.0177	-1.2033	-0.0379	-0.0045	0.7842	0.0038	-8.6685e-05	0.2411	0.0344	0.0099	-0.3157	-0.0035	3.0948e-04	-0.7844	-0.0077
2006141	-35.6287	-0.4284	-3.8690e-04	10.4069	0.1238	-0.0016	-0.7104	-0.0256	-9.4954e-04	0.8547	2.6275e-04	-0.0016	0.0559	-0.0063	1.8815e-05	-0.1830	-0.0022	0.0014	-0.7163	-0.0020
2006141	-36.1646	-0.4799	-0.0093	10.4515	0.1396	6.3749e-04	-0.6835	-0.0052	0.0019	0.8876	0.0148	5.8618e-04	0.3603	0.0192	0.0020	-0.0966	0.0130	0.0018	-0.7929	-0.0052

The project also explored the development of a custom function for feature extraction, which included the calculation of MFCCs, delta, and delta-delta coefficients from scratch, as well as the application of a Mel filterbank to the power spectral density of the voice signals. This custom approach was intended to deepen our understanding of the feature extraction process and potentially offer customized adjustments tailored to our dataset's specific needs. The function processed audio signals by breaking them into frames, applying window functions, and conducting Fourier transforms to extract frequency components, which were then filtered through a Mel filterbank to closely mimic the human ear's response.


The 26-filter Mel Filterbank
Amplitude
Frequency (Hz)
0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
0.8
0.9
1
0
1000
2000
3000
4000
5000
6000
7000
8000


Mel Spectrogram
Mel Filter Number
Frame Number
5
10
15
20
25
50
100
150
200
250
-20
-40
-60
-80
-100
-120
-140
-160
-180
-200



However, after careful consideration, we decided against incorporating the custom function into our final code. This decision was primarily influenced by the complexity and computational overhead introduced by the custom feature extraction process, which did not significantly outperform the built-in functions in terms of efficiency or accuracy. The built-in audio feature extractor provided a more streamlined and optimized solution, allowing for faster processing and simpler implementation without compromising the system's overall effectiveness. Moreover, the built-in functions were already validated and widely used in the field, ensuring reliability and reducing the potential for errors that might arise from custom-coded implementations.

```
Correlation between custom and built-in MFCCs: 0.050271
MSE between custom and built-in MFCCs: 437.3525
```

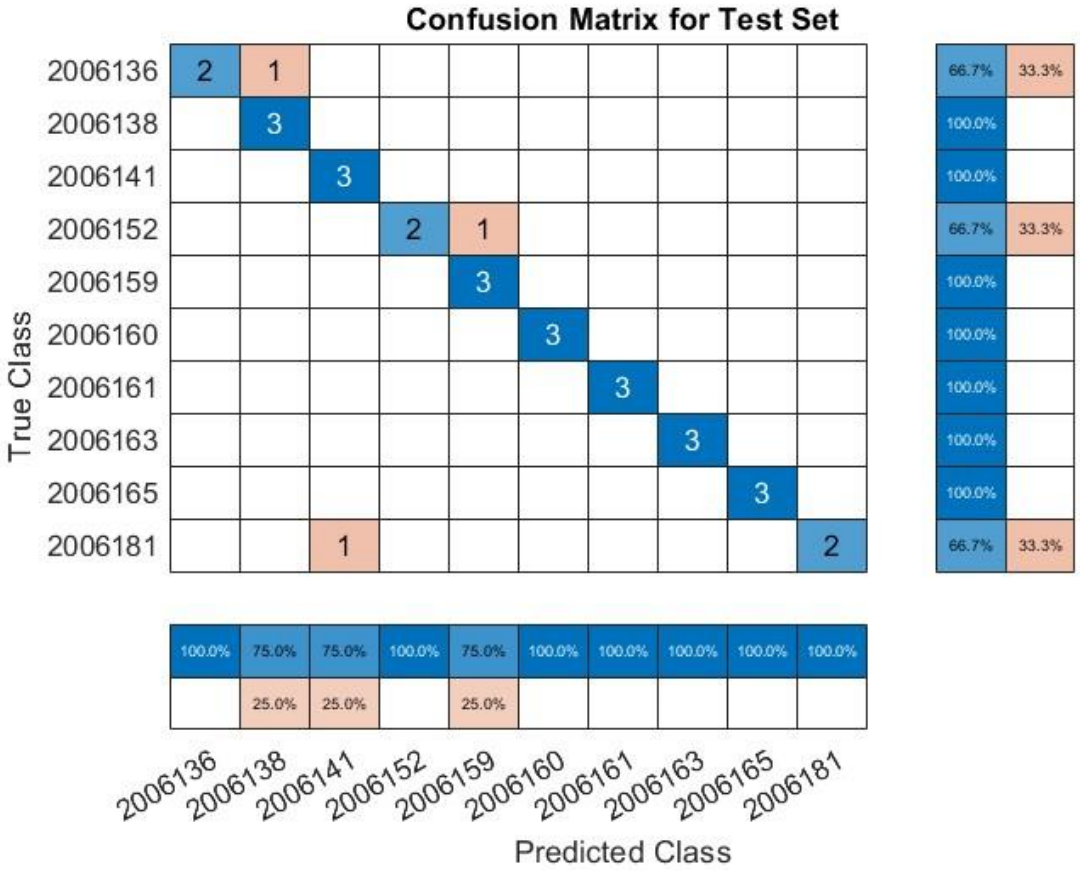
5.3.1.4 Training and Testing

In the development of our voice-based attendance system, an essential step was the careful division of our dataset into training and testing sets, ensuring a robust evaluation of our model's performance. We implemented an 80:20 split with 12 samples reserved for training and the remainder for testing. This approach aimed to maximize the diversity of voice data the model was exposed to during training, enhancing its ability to generalize across unseen data.

Following the dataset partitioning, we embarked on the model training and evaluation phase, employing a k-Nearest Neighbors (k-NN) classifier due to its efficacy in handling the complex feature spaces characteristic of voice data. Hyperparameter tuning across various

hyperparameters, including the number of neighbors (k) and distance metrics, was conducted to identify the optimal configuration that yielded the highest accuracy during a 5-fold cross-validation. This rigorous optimization process was crucial for fine-tuning the model, ensuring it could accurately classify voice samples based on the extracted features.

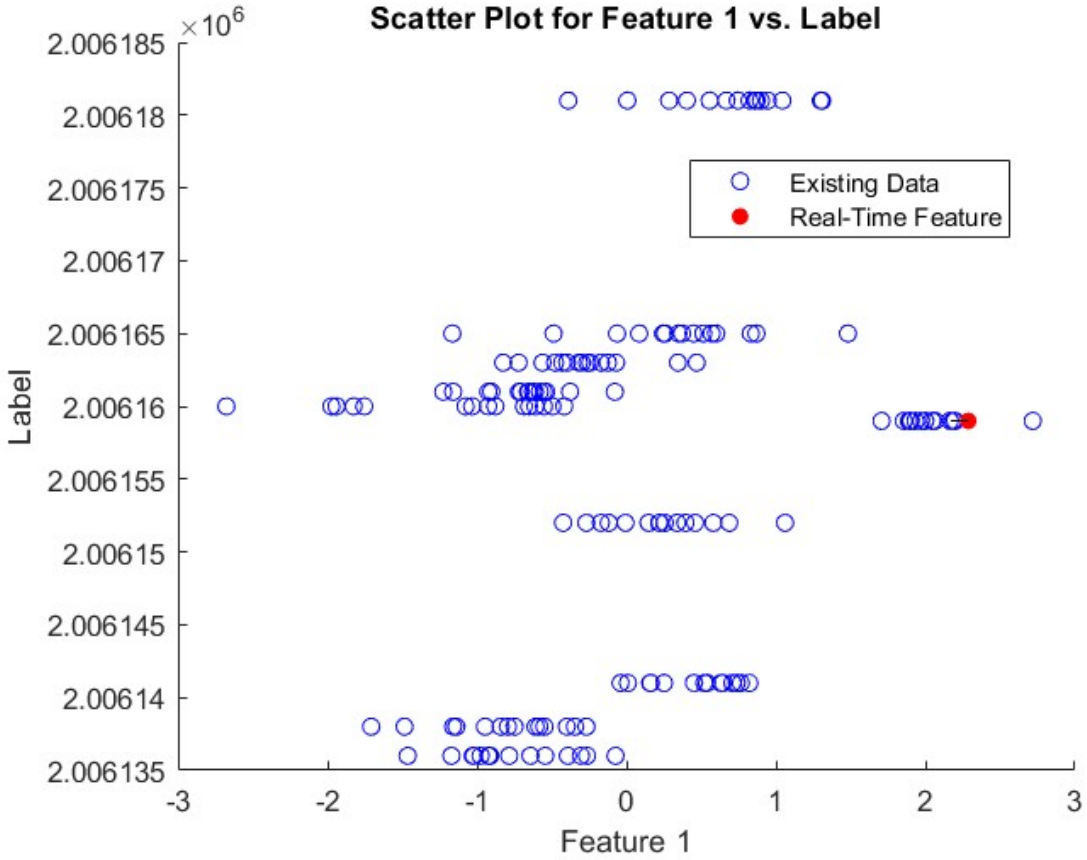
Once the optimal model parameters were determined, we evaluated the model's performance on the test set, applying the same normalization steps to maintain consistency in data processing. The results, illustrated through a confusion matrix, provided valuable insights into the model's predictive capabilities and areas for improvement. The final model, along with key parameters and normalization statistics, was saved for future application, marking a significant milestone in our project.



5.3.1.5 Real-time Testing

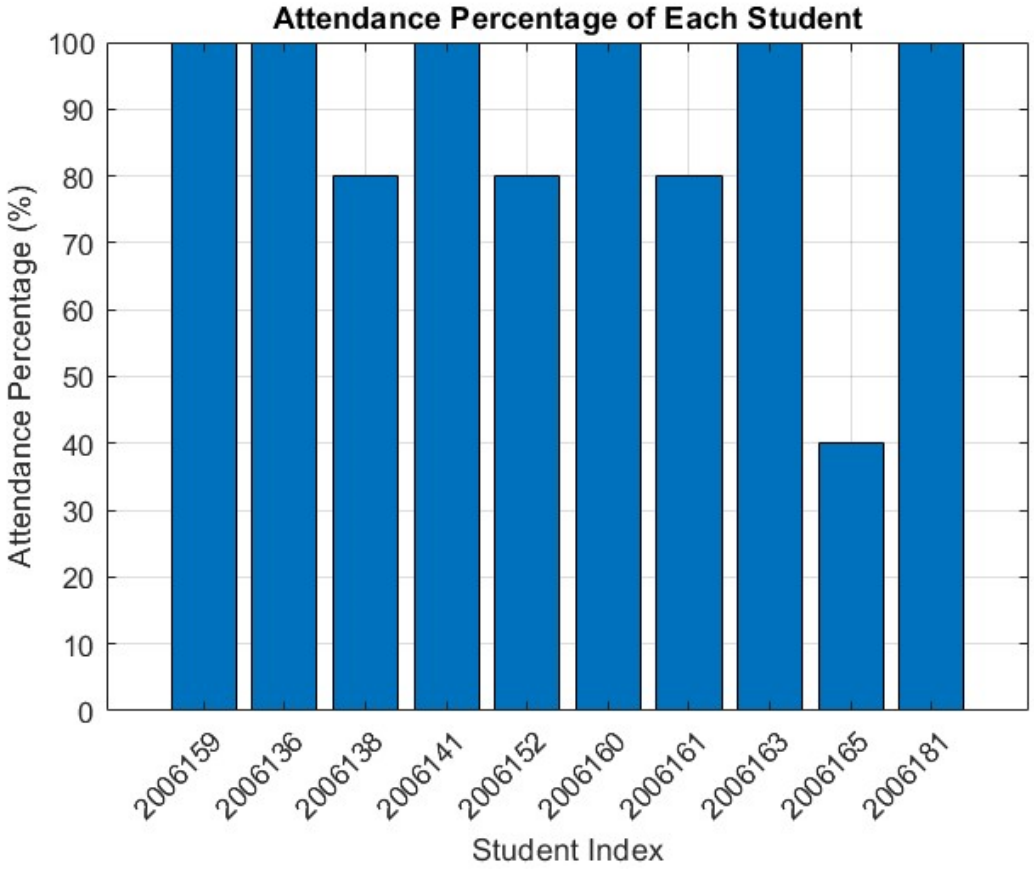
In the final stage of our voice-based attendance system, we implemented a real-time voice recording feature to capture and recognize individual student voices. Upon initiating the recording, the user's voice is captured for a duration of 3 seconds at a sampling frequency of 16,000Hz. This audio is then subjected to a series of preprocessing steps, including low-pass filtering to remove high-frequency noise, silence removal to focus on vocal content, and normalization to standardize the audio signal's amplitude. Following preprocessing, features are extracted using the previously trained audio feature extractor and normalized using the statistics (mean and standard deviation) derived from the training dataset. The processed features are then fed into the optimized k-Nearest Neighbors model, which predicts the student ID based on voice characteristics. The system evaluates the prediction's confidence by assessing the frequency of the most common label in relation to the total number of predictions, displaying the student ID if a sufficient level of confidence is achieved. This real-time processing and prediction mechanism not only demonstrates the system's ability to apply

learned patterns to new data but also emphasizes the practical application of our voice-based attendance system in educational settings.



5.3.1.6 Attendance Summary

Upon successfully identifying a student's voice using our trained model, the system updates an attendance database, marking the presence of the recognized student. This mechanism starts by checking if an `attendance_database.mat` file exists to load the current attendance records; if not, it initializes a new attendance table. For each identified student ID from the voice recognition phase, the system checks if the ID already exists in the database. If it's a new ID, the system adds a new entry with a presence mark ('P'); for existing IDs, it appends a 'P' to their attendance record. After updating, the database is saved, ensuring that each student's attendance history is accurately recorded. Furthermore, the system calculates and displays the attendance percentage for each student based on the total number of working days input by the user. This feature not only automates attendance tracking but also provides a transparent overview of attendance patterns, facilitating educational administration and enhancing the integrity of the attendance process.



5.3.1.7 Graphical User Interface

In our voice-based attendance system project, the final implementation stage involved developing a user-friendly graphical user interface (GUI) to facilitate easy interaction with the system. The GUI starts with a homepage featuring a "Get Started" button, intuitively guiding users into the application. Following this, a login page requires the user to enter a password, ensuring secure access to the system's functionalities. Upon successful authentication, the user is directed to a dashboard that serves as the central hub for all operations. This dashboard allows for the addition of new voice samples, enhancing the dataset's robustness and the system's accuracy. Users can also train the model directly from the dashboard, ensuring the system remains up-to-date with the latest data. Real-time attendance recording is a key feature, leveraging the system's voice recognition capabilities to efficiently track attendance without manual intervention. Additionally, an attendance summary can be viewed, offering insights into attendance trends and individual participation. This comprehensive GUI design ensures that the system is accessible and manageable, catering to the needs of both administrators and users, while maintaining a focus on security and efficiency.



5.3.2 Interpretation and Conclusions on Data

In conclusion, our comprehensive development of the biometric authentication system for classroom management demonstrates promising results in accurately recognizing and authenticating student voices. Through rigorous data collection, preprocessing, and feature extraction techniques, coupled with machine learning-based classification models, we have achieved satisfactory performance levels in identifying students based on their voice samples. Our systematic testing and validation procedures have confirmed the system's reliability and effectiveness across varying classroom conditions, including noisy environments. Additionally, the interpretation of our findings highlights the system's potential to enhance security and streamline attendance monitoring processes in educational settings. While further refinements and optimizations may be warranted to address certain limitations and enhance scalability, our research underscores the viability and practicality of biometric authentication technology in educational environments. Overall, our conclusion emphasizes the system's significant contributions towards fostering a safer and more efficient learning environment through advanced voice recognition capabilities.

5.4Limitations of Tools (PO(e))

Several limitations of the tools and technologies used in the development of the biometric authentication system are apparent:

- **Hardware Constraints:** The performance of our system may be constrained by the capabilities of the hardware utilized, particularly the processing power of the computer running the code and the quality of the microphone used for recording voice samples. These limitations could affect real-time processing and responsiveness, especially when dealing with large datasets or complex algorithms.
- **Software Dependencies:** Our reliance on specific software libraries and frameworks, such as MATLAB for signal processing and machine learning, may introduce limitations in terms of compatibility, support, and extensibility. Changes or updates to these dependencies could impact the functionality of our codebase and require modifications to ensure compatibility.
- **Algorithmic Constraints:** The effectiveness of our machine learning algorithms for voice recognition may be limited by factors such as the size and diversity of our training dataset, the complexity of feature extraction techniques implemented, and the chosen classification methods. Improvements in these areas are essential for enhancing the accuracy and reliability of our system.
- **Accuracy and Reliability:** Despite our best efforts, there may still be limitations in the accuracy and reliability of our voice recognition system, particularly in noisy environments or with speakers exhibiting variations in speech patterns. Addressing these limitations requires ongoing refinement and optimization of our algorithms and data processing techniques.
- **Scalability:** Our system's scalability may be limited by factors such as computational resources, data storage requirements, and the ability to handle large volumes of voice data efficiently. Optimizations in data processing and algorithmic efficiency are necessary to ensure scalability as our system grows in complexity and usage.

5.5Impact Assessment (PO(f))

5.5.1 Assessment of Societal and Cultural Issues

In the development of our biometric authentication system for classroom management, societal and cultural considerations are paramount. We assess the potential impact of the technology on diverse cultural contexts, ensuring sensitivity to cultural norms, values, and practices. Moreover, we evaluate the societal implications of implementing such a system, considering factors such as privacy concerns, data security, and the equitable treatment of individuals from various backgrounds.

5.5.2 Assessment of Health and Safety Issues

Health and safety considerations are integral to the implementation of our biometric authentication system. We assess the potential health risks associated with voice data collection and processing, ensuring compliance with relevant health and safety regulations. Additionally, we prioritize user safety by implementing measures to mitigate potential hazards, such as ergonomic considerations for system usage and adherence to recommended sound exposure limits during voice recording.

5.5.3 Assessment of Legal Issues

Legal compliance is a critical aspect of our biometric authentication system development. We conduct a thorough assessment of relevant laws and regulations governing data privacy, security, and biometric data usage. This includes compliance with international standards such as the General Data Protection Regulation (GDPR) and local regulations governing data protection and privacy. Additionally, we ensure transparency and accountability in data handling practices, providing users with clear information about their rights and the purposes for which their data is collected and processed.

5.6 Sustainability Evaluation (PO(g))

In assessing the sustainability of our biometric authentication system, we consider its environmental impact, resource utilization, and long-term viability. We evaluate the energy efficiency of the system's hardware components, aiming to minimize power consumption and reduce carbon emissions. Additionally, we assess the sustainability of our software architecture, prioritizing efficient algorithms and data processing techniques to optimize resource utilization. Furthermore, we explore strategies for extending the lifespan of the system through regular maintenance, updates, and adaptability to evolving technological standards. By prioritizing sustainability in our system design and implementation, we aim to minimize environmental harm and promote responsible stewardship of resources for future generations.

5.7 Ethical Issues (PO(h))

Ethical considerations are paramount in the development and deployment of our biometric authentication system. We uphold principles of fairness, transparency, and accountability throughout the system's lifecycle, ensuring that ethical guidelines and regulations are rigorously followed. We prioritize user privacy and consent, implementing robust security measures to protect sensitive biometric data from unauthorized access or misuse.

Additionally, we address ethical concerns related to algorithmic bias and discrimination, striving to mitigate biases and promote equitable treatment for all users. Furthermore, we engage stakeholders in ethical dialogue and decision-making processes, fostering a culture of ethical awareness and responsibility within our development team and broader community. By adhering to ethical principles, we aim to build trust, foster inclusivity, and uphold the dignity and rights of all individuals affected by our biometric authentication system.

6 Reflection on Individual and Teamwork (PO(i))

6.1 Individual Contribution of Each Member

Student ID	Contribution
2006138	Collected data, chose data collection parameters, and implemented real-time testing.
2006159	Worked on extracting important features, decided which features to focus on, and wrote custom functions for feature extraction.
2006160	Selected the best model for our project, implemented it, and also created a custom function for the model.
2006163	Handled data preprocessing, developed a custom function for detecting voice activity, and set up a database for attendance tracking.

7 Communication to External Stakeholders (PO(j))

7.1Executive Summary

Our project, based on a Voice-Based Attendance System, harnesses advanced signal processing and machine learning techniques for biometric authentication. Through meticulous data collection, training, and feature extraction from voice samples, the system ensures robust authentication while addressing ethical, societal, and legal considerations. Scalability and sustainability are emphasized, paving the way for real-world implementation in diverse educational settings. Future work involves refining accuracy, exploring alternative modalities, and fostering collaboration. In summary, our innovative solution promises streamlined attendance tracking, enhanced security, and optimized classroom engagement, representing a significant advancement in educational technology.

8 Future Work (PO(l))

In delineating future prospects, several strategic avenues emerge for the advancement and refinement of our biometric authentication system tailored for classroom management:

- **Enhanced Robustness and Accuracy:** Conduct research to fortify the system's robustness and accuracy by delving into advanced signal processing techniques and machine learning algorithms.
- **Exploration of Alternative Modalities:** Investigate alternative biometric modalities like facial or fingerprint recognition to complement existing voice recognition capabilities, thus broadening the scope of authentication options.
- **Scalability Optimization:** Address scalability challenges to accommodate larger student cohorts and diverse classroom environments, focusing on optimizing data handling and computational efficiency.
- **Personalization via Real-Time Feedback:** Integrate real-time feedback mechanisms and adaptive learning algorithms to personalize the authentication process, thereby optimizing performance for individual users.
- **Expansion of Functionality:** Extend the system's functionality to encompass additional classroom management tasks, including automated attendance tracking and student engagement monitoring, thereby enhancing overall educational efficiency.
- **Partnership Collaboration:** Foster collaborative partnerships with educational institutions and stakeholders to gain valuable insights and facilitate real-world implementation, ensuring alignment with practical classroom requirements.
- **Continuous Evaluation and Refinement:** Undertake ongoing evaluation and refinement initiatives grounded in user feedback and technological advancements to ensure sustained relevance and effectiveness in classroom management scenarios.

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